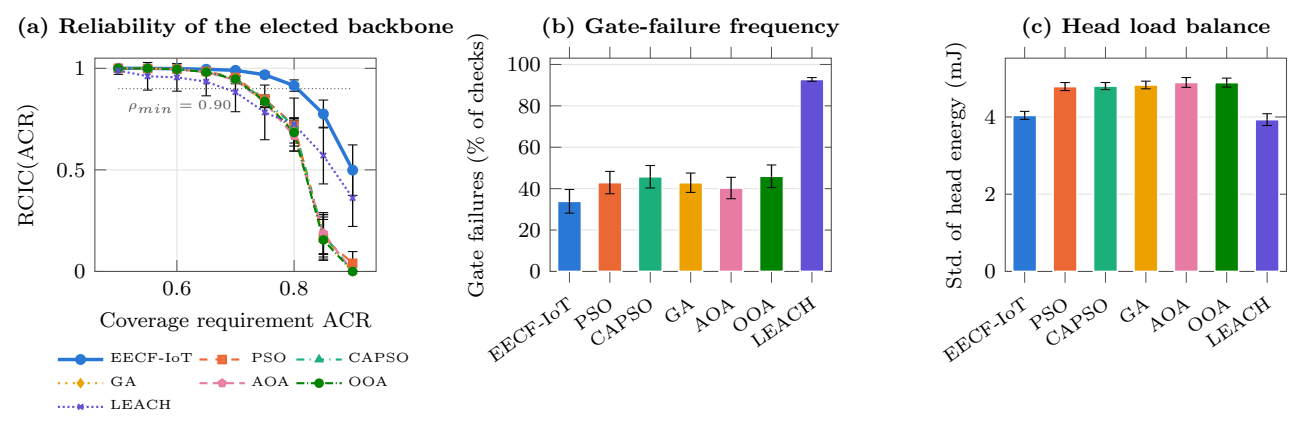

(a) Reliability of the elected backbone
RCIC(ACR)
1
0.5
0
0.6
0.8
$\rho_{min} = 0.90$
Coverage requirement ACR
EECF-IoT
PSO
CAPSO
GA
AOA
OOA
LEACH
(b) Gate-failure frequency
Gate failures (% of checks)
100
80
60
40
20
0
EECF-IoT
PSO
CAPSO
GA
AOA
OOA
LEACH
(c) Head load balance
Std. of head energy (mJ)
4
2
0
EECF-IoT
PSO
CAPSO
GA
AOA
OOA
LEACH